STAT 709 Mathematical Statistics Lecture 15 - Fall 2026

Lecture 15: Random vectors in high dimensions

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Priority guide

High	Core concepts to master for this lecture; will be directly examined.
Mid	Supporting concepts useful in other lectures; in scope of examinations.
Low	Enrichment supporting understanding; will not be examined.
Further reading	Optional self-study, references, extensions, and historical notes.

High priority *Overview*

Scalar concentration becomes a tool for vectors when we decide what to measure: all graph degrees, every covariance entry, a Euclidean norm, or finitely many pairwise distances. The recurring steps are to control one quantity, count the quantities of interest, and apply a union bound. The Gaussian and convex concentration statements, their matrix-norm example, and random projection are mid priority; the general concentration proofs and extensions remain optional.

High priority *Introduction*

In this lecture, we will continue studying concentration inequalities and introduce some fundamental high-dimensional phenomena. For a random vector $X = (X_1, \dots, X_n)$ with independent components, the previous lecture shows how to control $\langle a, X \rangle$ for a given nonrandom vector a.

As before, we use c or C to denote a positive absolute constant.

1 Examples of concentration inequalities

High priority *Examples of concentration inequalities*

Let us first recall concentration inequalities we learned in the previous lecture by looking at some examples.

Random graphs. The Erdős-Rényi graph $G \sim G(n, p)$ is a random graph with n vertices where we randomly draw an edge between vertex i and j with probability p. In other words, for $1 \le i < j \le n$, we have an independent Bernoulli random variable $A_{ij} \sim \text{Bern}(p)$. Set $A_{ji} = A_{ij}$ and $A_{ii} = 0$. The degree of vertex i is

$$d_i = \sum_{j \neq i} A_{ij}.$$

The expected degree for each i is $\mathbb{E}[d_i] = (n-1)p$, and we denote $d = (n-1)p$.

Proposition 1 (Dense graphs have concentrated degrees)**.** *There is an absolute constant C such that the following holds. Suppose the expected degree satisfies $d \geq C(\log n)^2$. Then, with probability*[1] *at least* 0.9 *all degrees of G are between* $0.9d$ *and* $1.1d$*.*

We can replace the constant number 0.9 to 0.99, and $0.9d$ and $1.1d$ closer to d (e.g., $0.99d$, $1.01d$). The same conclusion holds with a larger constant C.

Proof. To prove this inequality, we apply Bernstein's inequality for bounded random variables (Hoeffding's inequality is too crude): for any i,

$$\mathbb{P}\left(\left|d_i - d\right| \geq \frac{C_1(\sigma + M)}{\delta}\right) \leq 2\exp\left(-\min\left(\frac{1}{\delta}, \frac{1}{\delta^2}\right)\right), \qquad \text{for any } \delta > 0, \tag{1}$$

where C_1 is certain constant, $\sigma = (\mathrm{Var}(d_i))^{1/2}$ and $M \leq 1$ is a simple upper bound on $|A_{ij} - \mathbb{E}[A_{ij}]|$. Note $\sigma = ((n-1)p(1-p))^{1/2} \leq d^{1/2}$. We pick $\delta = (a \log n)^{-1}$ where $a = 6$ to get

$$\mathbb{P}\left(\left|d_i - d\right| \geq \frac{C_1(\sqrt{d}+1)}{\delta}\right) \leq 2\exp\left(-\min\left(\frac{1}{\delta}, \frac{1}{\delta^2}\right)\right) \leq 2n^{-a} \quad \text{when } a\log n \geq 1,$$

Note $C_1\frac{\sqrt{d}+1}{\delta} = O(\sqrt{d}\log n)$. Thus, under the assumption $d \geq C(\log n)^2$ for a large C, we have $C_1\frac{\sqrt{d}+1}{\delta} \leq 0.1d$. Thus, by the union bound,

$$\mathbb{P}\left(\exists\ i,\ \left|d_i - d\right| \geq 0.1d\right) \leq \sum_{i=1}^{n} \mathbb{P}\left(\left|d_i - d\right| \geq \frac{C_1(\sigma+M)}{\delta}\right) \leq 2n^{1-a}$$

For $n \geq 2$ and $a = 6$, the right-hand side is at most $2 \cdot 2^{-5} < 0.1$; enlarging the constant in the degree assumption ensures the radius comparison above. □

The coarser inverted radius used above is not optimal. The direct bounded Bernstein inequality, or a multiplicative Chernoff inequality, lets us replace $(\log n)^2$ by $\log n$ in the statement of Proposition 1. See [2][Prop. 2.4.1] for details.

Covariance matrix. Let $y_1, \ldots, y_n \overset{\text{i.i.d.}}{\sim} Y \in \mathbb{R}^d$ be i.i.d. random vectors with zero mean and subgaussian components, namely $\mathbb{E}\,Y = 0$ and $\max_{1\leq j\leq d} \|Y_j\|_{\psi_2} \leq K$. All coordinate products are integrable: Cauchy–Schwarz gives $\mathbb{E}\,|Y_jY_m| \leq (\mathbb{E}\,Y_j^2\,\mathbb{E}\,Y_m^2)^{1/2} < \infty$. Thus the covariance matrix exists and is denoted by $\Sigma = \mathrm{Cov}(Y)$ which is a $d \times d$ matrix. The empirical covariance matrix is

$$\widehat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n} y_i y_i^\top$$

We can apply Bernstein's inequality to bound the estimation error under the max norm:

$$\left\|\widehat{\Sigma} - \Sigma\right\|_{\max} := \max_{1\leq j,m\leq d} \left|\widehat{\Sigma} - \Sigma\right|_{j,m} = \max_{1\leq j,m\leq d} \left|\frac{1}{n}\sum_{i=1}^{n} \left([y_i]_j[y_i]_m - \mathbb{E}[[y_i]_j[y_i]_m]\right)\right|.$$

For each fixed pair (j, m), the summands are independent across observations i. Coordinates within an observation need not be independent. The product and centering bounds give $\|Y_jY_m - \mathbb{E}(Y_jY_m)\|_{\psi_1} \leq 4K^2$, so Bernstein's inequality yields

$$P\left(|[\widehat{\Sigma} - \Sigma]_{jm}| \geq tK^2\right) \leq 2\exp\left(-cn\min(t^2, t)\right).$$

[1]Here the theorem gives the stated fixed confidence of 0.9. In an asymptotic statement, "with high probability" usually means probability tending to one; that stronger statement requires a corresponding choice of the constants or growth assumptions.

We choose $t = C\sqrt{(\log d)/n}$ with a sufficiently large $C \geq 1$ so that $cnt^2 \geq 4\log d$. Assume $d \geq 2$ and $n \geq C^2 \log d$, so $t \leq 1$ so $\min(t^2, t) = t^2$. This yields

$$P\left(|[\widehat{\Sigma} - \Sigma]_{jm}| \geq CK^2\sqrt{\frac{\log d}{n}}\right) \leq 2\exp(-4\log d) = 2d^{-4}.$$

By applying a union bound over the choice of j, m, we get

$$P\left(\left\|\widehat{\Sigma} - \Sigma\right\|_{\max} \geq CK^2\sqrt{\frac{\log d}{n}}\right) \leq 2d^{-2}.$$

Note that by adjusting the constant, we can make the exponent -2 in d^{-2} smaller (like -10, -100, etc.)

Uniform control. A union bound converts one-entry concentration into simultaneous control of graph degrees or covariance entries. The number of quantities contributes a logarithmic factor through the chosen failure probability.

High priority *Practice: A sharper degree requirement*

Exercise prompt. Use bounded Bernstein directly to show that for $0 < \eta < 1$, $\mathbb{P}(\max_i |d_i - d| \geq \eta d) \leq 2n\exp[-\eta^2 d/(2(1+\eta/3))]$.

See Exercise 1 in Section 7 for the full statement and complete solution.

2 Two general concentration inequalities

Low priority *Two general concentration inequalities*

It is possible to generalize the concentration inequalities beyond a linear combination of independent random variables. The following result is a classic concentration inequality for general **Lipschitz** functions, although it only applies to Gaussian random variables instead of subgaussian random variables.

Mid priority *Gaussian concentration inequality*

Theorem 1 (Gaussian concentration inequality)**.** *Let* $X_1, X_2, \ldots X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. *Let* $F : \mathbb{R}^n \to \mathbb{R}$ *be an* L*-Lipschitz function, i.e.,* $|F(x) - F(y)| \leq L\|x - y\|$. *Then*

$$\|F(X_1, \ldots, X_n) - \mathbb{E}F(X_1, \ldots, X_n)\|_{\psi_2} \leq CL$$

where C *is a constant.*

Low priority *Gaussian concentration: proof and details*

If we consider the special case $F(x) = \langle a, x\rangle$, then F is $\|a\|$-Lipschitz. This leads to the same bound as in Hoeffding's inequality.

Proof of Theorem 1.* For simplicity, denote $X = (X_1, \ldots, X_n)$, and let Y be an independent copy. First prove the claim for smooth F with $\|\nabla F\| \leq L$; convolution approximation extends it to every Lipschitz F, with dominated convergence justified by linear growth and Gaussian exponential integrability. We also introduce the definition $X_\theta = X\cos\theta + Y\sin\theta$. Using Markov's inequality and Jensen's inequality, we obtain

$$P(F(X) - \mathbb{E}\,F(X) > Lt) \leq e^{-\lambda Lt}\,\mathbb{E}\,e^{\lambda(F(X) - \mathbb{E}\,F(X))} \leq e^{-\lambda Lt}\,\mathbb{E}\,e^{\lambda(F(X) - F(Y))}.$$

Note that

$$\frac{d}{d\theta}X_\theta = -X\sin\theta + Y\cos\theta \sim N(0, I_n)$$

for any $\theta \in [0, \pi/2]$. A crucial observation is that $\frac{d}{d\theta}X_\theta$ and X_θ are independent, since $(X_\theta, \frac{d}{d\theta}X_\theta)$ is orthogonal transform of X, Y. Denote $X'_\theta = \frac{d}{d\theta}X_\theta$.

We use Jensen's inequality for the integration (viewed as measure integration w.r.t. Unif$[0, \pi/2]$) in the exponentiation and Fubini's theorem,

$$\begin{aligned}
\mathbb{E}\, e^{\lambda[F(Y)-F(X)]} &= \mathbb{E}\exp\Big(\frac{\pi\lambda}{2}\int_0^{\pi/2}(-X\sin\theta + Y\cos\theta)^T\nabla F(X_\theta)\frac{2}{\pi}d\theta\Big) \\
&\le \frac{2}{\pi}\int_0^{\pi/2}\mathbb{E}\, e^{\frac{\pi\lambda}{2}(-X\sin\theta+Y\cos\theta)^T\nabla F(X_\theta)}d\theta \qquad \text{(by Jensen and Fubini)} \\
&= \frac{2}{\pi}\int_0^{\pi/2}\mathbb{E}\, e^{\frac{\pi\lambda}{2}(X'_\theta)^T\nabla F(X_\theta)}d\theta \\
&= \frac{2}{\pi}\int_0^{\pi/2}\mathbb{E}e^{\frac{\lambda^2\pi^2\|\nabla F(X_\theta)\|^2}{8}}d\theta \qquad \text{(by independence)} \\
&\le e^{\frac{\lambda^2\pi^2L^2}{8}}. \qquad \text{(by Lipschitz assumption)}
\end{aligned}$$

The difference $F(X) - F(Y)$ has the same distribution as $F(Y) - F(X)$ by exchanging the independent copies. The preceding bound therefore applies to the MGF in the Markov step. If $L = 0$, F is constant. Otherwise choose $\lambda = \frac{4t}{\pi^2 L}$ to obtain

$$P(F(X) - \mathbb{E}\,F(X) > Lt) \le e^{-\frac{2t^2}{\pi^2}}.$$

Applying the same argument to $-F(X)$, we can derive that

$$P(-F(X) + \mathbb{E}\,F(X) > Lt) \le e^{-\frac{2t^2}{\pi^2}},$$

and therefore

$$P(|F(X) - \mathbb{E}\,F(X)| > Lt) \le 2e^{-\frac{2t^2}{\pi^2}}.$$

□

Mid priority *Concentration of Gaussian matrix norms*

Example 1. *Suppose that Y is a matrix of i.i.d. standard normal entries. Let $\|\cdot\|$ be either the Frobenius norm or the matrix operator norm. Then,*

$$P\big(\big|\|Y\| - \mathbb{E}\|Y\|\big| > t\big) \le 2e^{-ct^2}, \qquad \textit{for all } t > 0.$$

Low priority *Matrix norms: explanation*

The above inequality follows from Theorem 1 and the observation that $|f(Y) - f(\widetilde{Y})| \le f(Y - \widetilde{Y})$ with $f(Y) = \|Y\|$ (due to triangle inequality in the definition of a norm). Remarkably, the deviation of $\|Y\|$ to its expectation is $O(1)$, which does not depend on the dimensions of the matrix Y. As we will learn in the next lecture, $\mathbb{E}\|Y\|$ generally scales with the matrix dimension. In the special case of vectors $Y \in \mathbb{R}^n$, from homework problems we have learned $\|Y\| = \sqrt{n} + O_P(1)$ (see also next section), so indeed $\|Y\|$ concentrates around its mean.

What if we don't have Gaussian distributions? Talagrand's concentration inequality.

Mid priority *Talagrand's concentration inequality*

Theorem 2 (Talagrand's concentration inequality)**.** *Suppose that $X_1, \dots, X_n$ are independent random variables satisfying $\max_i |X_i| \le K$. Denote $X = (X_1, \dots, X_n)$. Let $F : \mathbb{R}^n \to \mathbb{R}$ be a **convex** L-Lipschitz function. Then*

$$P\big(|F(X) - \mathbb{E}[F(X)]| \ge tL\big) \le C \exp\Big(-\frac{ct^2}{K^2}\Big) \qquad \text{for any } t > 0. \tag{2}$$

Low priority *Talagrand concentration: proof and extension*

See [1, Theorem 2.1.13] for a proof. This theorem says that for a bounded random vector with independent coordinates, $F(X) - \mathbb{E}[F(X)]$ is a subgaussian with norm bounded by $O(KL)$.

In place of Gaussianity, we now require bounded independent coordinates and convexity. For independent variables ξ_i with $\max_i \|\xi_i\|_{\psi_2} \le K$, one practical extension uses truncation:

$$X_i = \xi_i \mathbf{1}\{|\xi_i| \le \tau_n\}$$

at appropriate threshold τ_n. For example, choose $\tau_n = C_q K \sqrt{\log n}$ for subgaussian ξ_i allows us to have $P(X_i = \xi_i, \forall\ i) \ge 1 - C' n^{-q}$ for any $q > 0$. For convex F, we may then apply Talagrandś concentration inequality to X_i, obtain the concentration inequality of $F(X)$, and control the change in center using

$$|\,\mathbb{E}\,F(X) - \mathbb{E}\,F(\xi)| \le L\,\mathbb{E}\,\|X - \xi\|_2 \le L\Big(\sum_i \mathbb{E}[\xi_i^2 \mathbf{1}\{|\xi_i| > \tau_n\}]\Big)^{1/2}.$$

Integrating the subgaussian tail makes this small by increasing C_q. Together with the truncation failure probability, this transfers the bound to $F(\xi)$, at the cost of the logarithmic scale in τ_n.

Low priority *Practice: The norm is Lipschitz*

Exercise prompt. Show that $x \mapsto \|x\|_2$ and $A \mapsto \|A\|_{\mathrm{op}}$ are 1-Lipschitz with respect to Euclidean and Frobenius distance, respectively.

See Exercise 2 in Section 7 for the full statement and complete solution.

3 Norm of random vectors

High priority *Norm of random vectors*

Suppose that $X = (X_1, \dots, X_n)$ is a random vector where $X_1, \dots, X_n$ are independent random variables with $\mathbb{E}[X_i] = 0$ and $\mathrm{Var}(X_i) = 1$. We are interested in the Euclidean norm $\|X\|$. Let's gather some results about $\|X\|$ we have proved so far.

1. For i.i.d. coordinates, by the LLN and continuous mapping theorem, $n^{-1/2}\|X\| \xrightarrow{p} 1$ as $n \to \infty$.
2. For i.i.d. coordinates with finite fourth moment, the CLT and delta method give $\|X\| - \sqrt{n} \Rightarrow N(0, \mathrm{Var}(X_1^2)/4)$. For standard Gaussian coordinates, this variance is $1/2$.
3. By Jensen's inequality, $\mathbb{E}[\|X\|] \le \sqrt{\mathbb{E}[\|X\|^2]} = \sqrt{n}$.
4. If X_i has distribution $N(0,1)$ or $\max_i |X_i| \le C$, then $\|X\| - \mathbb{E}[\|X\|]$ has subgaussian norm $O(1)$.

Points 1 and 2 suggest that $\|X\|$ concentrates around $\sqrt{n}$, but do not supply a finite-sample tail bound. Point 4 follows from the preceding mid-priority concentration statements; the direct Bernstein proof below supplies an independent main-route argument. Theorem 3.1.1 of [2] shows the following result.

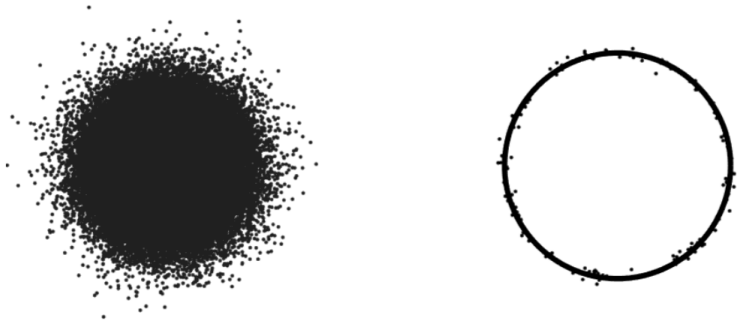


Figure 3.6 A Gaussian point cloud in two dimensions (left) and its intuitive visualization in high dimensions (right). In high dimensions, the standard normal distribution is very close to the uniform distribution on the sphere of radius $\sqrt{n}$.

Theorem 3. *Let $X = (X_1, \dots, X_n)$ be a random vector where $X_1, \dots, X_n$ are independent random variables with $\mathbb{E}(X_i^2) = 1$ and $\max_i \|X_i\|_{\psi_2} \le K$. Then,*

$$\big\| \|X\|_2 - \sqrt{n} \big\|_{\psi_2} \le CK^2.$$

Proof. For convenience, we assume $K \ge 1$. Let $Y_i = X_i^2 - 1$. The product and centering inequalities from Lecture 14 give $\|Y_i\|_{\psi_1} \le 2\|X_i^2\|_{\psi_1} \le 4K^2$. Applying Bernstein's inequality,

$$P\left(\Big| \frac{1}{n}\|X\|^2 - 1 \Big| \ge u \right) \le 2\exp\left(-\frac{cn}{K^4} \min(u, u^2) \right), \qquad \text{for all } u > 0 \tag{3}$$

where we used $K^4 \ge K^2$ due to the assumption $K \ge 1$. To prove the tail bound for $\|X\|$, we use the following observation:

$$|z - 1| \ge \delta \quad \Longrightarrow \quad |z^2 - 1| \ge \max(\delta, \delta^2), \qquad \text{for all } z \ge 0,\ \delta > 0$$

Thus we obtain

$$P\left(\Big| \frac{1}{\sqrt{n}}\|X\| - 1 \Big| \ge \delta \right) \le P\left(\Big| \frac{1}{n}\|X\|^2 - 1 \Big| \ge \max(\delta, \delta^2) \right) \le 2\exp\left(-\frac{cn}{K^4}\delta^2 \right)$$

where we used $\min(u, u^2) \ge \delta^2$ with $u = \max(\delta, \delta^2)$. Changing variables to $t = \delta\sqrt{n}$, we therefore obtain

$$P\big(\big| \|X\| - \sqrt{n} \big| \ge t \big) \le 2\exp\left(-\frac{ct^2}{K^4} \right) \qquad \text{for all } t \ge 0.$$

□

Uniform spherical random variable. Recall that we introduced the uniform random variable on the sphere $\mathrm{Unif}(\mathbb{S}^{n-1})$ in the n-dimensional Euclidean space. We use $r\mathbb{S}^{n-1}$ to denote the sphere in $\mathbb{R}^n$ with radius r.

Theorem 3 gives a useful radial coupling between a Gaussian vector and a uniform point on the sphere of radius $\sqrt{n}$. In fact, if $G \sim N(0, I_n)$, then

$$G = UR, \qquad \text{where } U = G/\|G\|,\ R = \|G\|$$

satisfies $U \sim \mathrm{Unif}(\mathbb{S}^{n-1})$ and U and R are independent. Since R concentrates around $\sqrt{n}$, the probability mass of G is mostly around a thin spherical layer of radius $\sqrt{n}$. The absolute shell width stays of order one while its radius grows like $\sqrt{n}$, so the relative width tends to zero. The Gaussian and sphere distributions are nevertheless different: one has a density in space, while the other lives on a sphere.

Theorem 4 (Uniform spherical distribution is subgaussian)**.** *Denote the inner product of vectors by* $\langle u, v\rangle = u^\top v$*. The random vector* $X \sim \mathrm{Unif}(\sqrt{n}\,\mathbb{S}^{n-1})$ *satisfies*

$$\sup_{u\in\mathbb{R}^n:\|u\|=1} \|\langle X, u\rangle\|_{\psi_2} \le C\,. \tag{4}$$

Namely, projecting X *using any unit vector gives a subgaussian random variable with a norm bounded by* C*.*

This result is intuitively true because of the relation

$$X = \sqrt{n}\frac{G}{\|G\|}$$

Since $\|G\| \approx \sqrt{n}$ and $\langle G, u\rangle \sim N(0,1)$ for any choice of unit vector u, we expect (4) is true.

Proof sketch. A rigorous proof is based on defining a "high probability event"

$$\mathcal{E} = \left\{\|G\| \ge \frac{\sqrt{n}}{2}\right\}$$

which satisfies $P(\mathcal{E}^c) \le 2\exp(-cn)$. So

$$\begin{aligned} P\left(|\langle X, u\rangle| \ge t\right) &\le P\left(\frac{|\langle G, u\rangle|}{\|G\|} \ge \frac{t}{\sqrt{n}} \text{ and } \mathcal{E}\right) + P(\mathcal{E}^c) \\ &\le P\left(|\langle G, u\rangle| \ge \frac{t}{2} \text{ and } \mathcal{E}\right) + 2\exp(-cn) \\ &\le P\left(|\langle G, u\rangle| \ge \frac{t}{2}\right) + 2\exp(-cn) \\ &\le 2\exp(-t^2/8) + 2\exp(-cn). \end{aligned}$$

For $t \le \sqrt{n}$, the right side is at most $4e^{-c_1 t^2}$. Combining this with the trivial probability bound 1 and reducing the absolute constant gives $2e^{-c_2 t^2}$, the required subgaussian form. For $t > \sqrt{n}$, the probability is trivially zero since $|\langle X, u\rangle| \le \|X\| = \sqrt{n}$. □

Instructor's Note *Most Gaussian points live in a shell*

The origin is the peak of the Gaussian density, but density at one location is not the same as probability in a region. There are many more directions and much more volume away from the origin. In high dimension that balance puts most points in a thin shell of radius about $\sqrt{n}$, even though each coordinate is still ordinary Gaussian noise.

Radial concentration. For independent isotropic subgaussian coordinates, the Euclidean norm is close to $\sqrt{n}$, even though each coordinate fluctuates. This radial statement differs from concentration of a fixed coordinate.

High priority *Practice: Norm fluctuations for random signs*

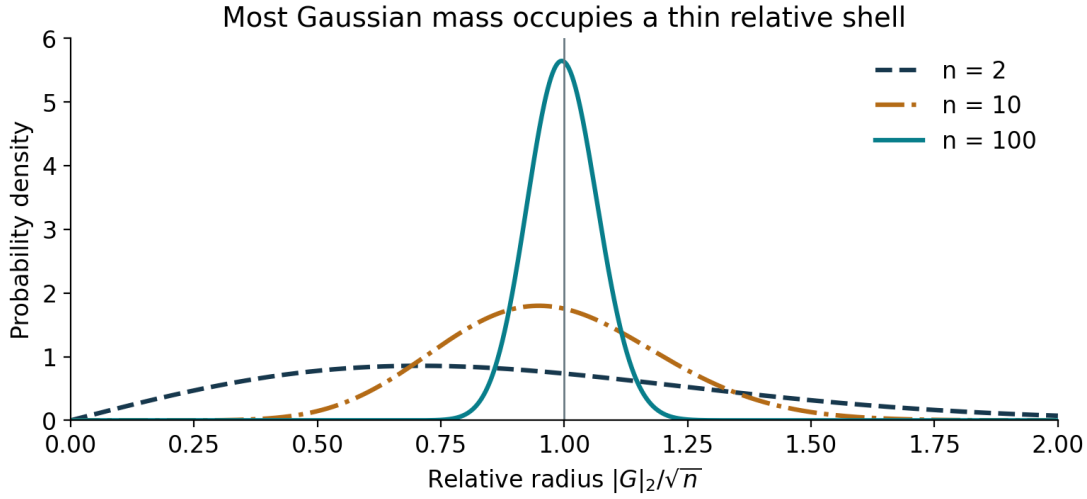


Figure 1: For $G \sim N(0, I_n)$, the relative radius $\|G\|_2/\sqrt{n}$ concentrates near one. The curves are exact radial densities; the thin-shell statement concerns probability mass, not the density at a single point.

Exercise prompt. If the coordinates are independent signs taking values ± 1, what are $\|X\|_2$ and the limiting variance in the norm delta method?

See Exercise 3 in Section 7 for the full statement and complete solution.

4 Random projection

Mid priority *Random projection*

If we have a large collection of high-dimensional points $Q := \{x_i\}_{i \le N} \subset \mathbb{R}^d$, we can reduce dimensionality if the points are approximately in a low-dimensional subspace. A surprising result states that for an arbitrary collection of points, we can randomly map the points to a low-dimensional space while approximately preserving the pairwise distances.

Theorem 5 (Johnson-Lindenstrauss lemma)**.** *Let $\varepsilon \in (0, 1/2)$. There exists a constant C such that the following statement is true. Let $k \ge \frac{C \log N}{\varepsilon^2}$ be an integer. Then there exists a linear mapping $f : \mathbb{R}^d \to \mathbb{R}^k$ such that*

$$(1-\varepsilon)\|x-y\|^2 \le \|f(x)-f(y)\|^2 \le (1+\varepsilon)\|x-y\|^2, \qquad \text{for all } x, y \in Q$$

This result is useful when d is much larger than $O((\log N)/\varepsilon^2)$.

Normalization. The Gaussian map $f(x) = Ax/\sqrt{k}$ satisfies $\mathbb{E}\,\|f(x)-f(y)\|^2 = \|x-y\|^2$ for every fixed pair. Concentration and a union bound then control all $N(N-1)/2$ pairwise distances simultaneously.

Proof. Let Q be fixed before drawing $A \in \mathbb{R}^{k \times d}$, whose entries are independent with $A_{ij} \overset{\text{i.i.d.}}{\sim} N(0,1)$. Our goal is to show

$$P\left((1-\varepsilon)\|u\|^2 \le \left\|\frac{1}{\sqrt{k}}Au\right\|^2 \le (1+\varepsilon)\|u\|^2\right) \ge 1 - 2\exp\left(-c\varepsilon^2 k\right), \qquad \text{for any } u \in \mathbb{R}^d. \tag{5}$$

Once this is proved, then we apply this inequality to $u = x - y$ for every pair $x, y \in Q$. By the union bound,

$$P\left(\forall x, y \in Q,\ (1-\varepsilon)\|x-y\|^2 \le \left\|\frac{1}{\sqrt{k}}A(x-y)\right\|^2 \le (1+\varepsilon)\|x-y\|^2\right) \ge 1 - 2N^2 \exp\left(-c\varepsilon^2 k\right)$$

The right-hand side is larger than $1 - O(N^{-2})$ after choosing the absolute constant sufficiently large under our assumption. To show Eqn. 5, we denote $z = Au/\sqrt{k}$. It is easy to check (by independence of the Gaussian rows) that $z \sim N(0, (\|u\|^2/k) I_k)$ with independent coordinates. Each row of A has inner product $N(0, \|u\|^2)$ with u, and distinct rows are independent. Thus, our goal Eqn. 5 is the same as

$$P\Big(\big|\|z\|^2 - \mathbb{E}\,\|z\|^2\big| \le \varepsilon\|u\|^2\Big) \ge 1 - 2\exp\left(-c\varepsilon^2 k\right).$$

If $u = 0$, the success event has probability one. We can assume without loss of generality that $\|u\| = 1$ since we can always consider rescaling $z/\|u\|$. By Bernstein's inequality in Eqn. (3), we get exactly the above desired inequality. □

Finite-set dimension reduction. A Gaussian projection preserves the squared lengths of all pairwise differences in a finite set after a union bound. The required target dimension grows logarithmically with the number of points and quadratically in inverse distortion.

Mid priority *Practice: Count the required projections*

Exercise prompt. If every fixed difference vector has distortion failure probability at most $2e^{-c\epsilon^2 k}$, give a sufficient k for $N \ge 2$ fixed points to have all pairwise distances preserved with probability at least $1 - \alpha$.

See Exercise 4 in Section 7 for the full statement and complete solution.

5 Further reading

Further reading *Further reading*

Section 2 follows Theorems 2.1.12–2.1.13 of [1]; the author's text includes the proofs. The applications, norm concentration, sphere geometry, and random projection follow [2, Sections 2.4, 3.1, 3.4, 5.3].

6 Key ideas to retain

High priority *Uniform control*

A union bound converts one-entry concentration into simultaneous control of graph degrees or covariance entries. The number of quantities contributes a logarithmic factor through the chosen failure probability. See the source checkpoint.

High priority *Radial concentration*

For independent isotropic subgaussian coordinates, the Euclidean norm is close to $\sqrt{n}$, even though each coordinate fluctuates. This radial statement differs from concentration of a fixed coordinate. See the source checkpoint.

Mid priority *Finite-set dimension reduction*

A Gaussian projection preserves the squared lengths of all pairwise differences in a finite set after a union bound. The required target dimension grows logarithmically with the number of points and quadratically in inverse distortion. See the source checkpoint.

7 Exercises and complete solutions

High priority *A sharper degree requirement*

Exercise 1 (A sharper degree requirement).
Use bounded Bernstein directly to show that for $0 < \eta < 1$, $\mathbb{P}(\max_i |d_i - d| \geq \eta d) \leq 2n \exp[-\eta^2 d/(2(1+\eta/3))]$.

Solution.
For fixed i, $d_i - d$ sums independent centered Bernoulli terms with total variance at most d and absolute bound 1. Substitute $t = \eta d$ into Bernstein to get the exponential factor. A union bound over n vertices gives the assertion; degrees need not be mutually independent. Thus $d \geq C_\eta \log(n/\alpha)$ suffices for failure probability at most α.

□

Low priority *The norm is Lipschitz*

Exercise 2 (The norm is Lipschitz).
Show that $x \mapsto \|x\|_2$ and $A \mapsto \|A\|_{\mathrm{op}}$ are 1-Lipschitz with respect to Euclidean and Frobenius distance, respectively.

Solution.
The reverse triangle inequality gives $|\|x\| - \|y\|| \leq \|x - y\|$. Likewise $|\|A\|_{\mathrm{op}} - \|B\|_{\mathrm{op}}| \leq \|A - B\|_{\mathrm{op}} \leq \|A - B\|_F$, since the largest squared singular value is at most their sum. Both functions are also convex.

□

High priority *Norm fluctuations for random signs*

Exercise 3 (Norm fluctuations for random signs).
If the coordinates are independent signs taking values ± 1, what are $\|X\|_2$ and the limiting variance in the norm delta method?

Solution.
Each square is 1, so $\|X\|_2 = \sqrt{n}$ exactly. Also $\mathrm{Var}(X_1^2) = 0$, giving a degenerate limiting Normal with variance zero. This demonstrates why the Gaussian variance $1/2$ is not a universal norm fluctuation.

□

Mid priority *Count the required projections*

Exercise 4 (Count the required projections).
If every fixed difference vector has distortion failure probability at most $2e^{-c\epsilon^2 k}$, give a sufficient k for $N \geq 2$ fixed points to have all pairwise distances preserved with probability at least $1 - \alpha$.

Solution.
There are at most $N(N-1)/2$ unordered pairs. The failure probability is at most $N(N-1)e^{-c\epsilon^2 k} \leq N^2 e^{-c\epsilon^2 k}$. Hence $k \geq [2 \log N + \log(1/\alpha)]/(c\epsilon^2)$ suffices. The point set is fixed independently of the random map.

□

High priority *All covariance entries at once*

Exercise 5 (All covariance entries at once).

Under the coordinatewise subgaussian assumption, derive the max-norm covariance rate without assuming $\log d \le n$.

Solution.

Products have ψ_1 norm at most $2K^2$, and centered products at most $4K^2$. Bernstein and a union bound over d^2 entries give $2d^2 \exp[-cn \min(t^2, t)]$ for threshold $CK^2 t$. Taking $t = C \max\{\sqrt{u/n}, u/n\}$ with $u = \log(2d^2/\alpha)$ yields failure at most α for a sufficiently large absolute constant. This retains the linear-deviation regime.

□

High priority *The event comparison for a square root*

Exercise 6 (The event comparison for a square root).

For $z \ge 0$, prove $|z - 1| \ge \delta$ implies $|z^2 - 1| \ge \max(\delta, \delta^2)$.

Solution.

If $z \ge 1+\delta$, then $z^2 - 1 \ge 2\delta + \delta^2$. If $0 \le z \le 1-\delta$, necessarily $\delta \le 1$, and $1 - z^2 \ge 2\delta - \delta^2 \ge \delta = \max(\delta, \delta^2)$. For $\delta > 1$ the second case is impossible.

□

High priority *Covariance of a uniform direction*

Exercise 7 (Covariance of a uniform direction).

For $U \sim \mathrm{Unif}(\mathbb{S}^{n-1})$, prove $\mathbb{E}\, U = 0$ and $\mathbb{E}\, UU^\top = I_n/n$.

Solution.

Reflecting one coordinate preserves the law, so each mean and each off-diagonal product expectation is zero. Permuting coordinates preserves the law, so all diagonal second moments are equal. Their sum is $\mathbb{E}\,\|U\|^2 = 1$, giving $1/n$ each. Thus $\sqrt{n}U$ has identity covariance despite dependent coordinates.

□

Mid priority *A general norm central limit theorem*

Exercise 8 (A general norm central limit theorem).

For i.i.d. mean-zero unit-variance coordinates with finite fourth moment, prove the norm limit with variance $\mathrm{Var}(X_1^2)/4$.

Solution.

Put $Q_n = n^{-1} \sum X_i^2$. The CLT gives $\sqrt{n}(Q_n - 1) \Rightarrow N(0, \mathrm{Var}(X_1^2))$. Since $g(q) = \sqrt{q}$ has derivative $1/2$ at 1, $\sqrt{n}(\sqrt{Q_n} - 1) = \|X\| - \sqrt{n}$ converges to the asserted law, including the zero-variance case.

□

References

[1] Terence Tao. *Topics in random matrix theory*, volume 132. American Mathematical Soc., 2012.

[2] Roman Vershynin. *High-dimensional probability: An introduction with applications in data science*, volume 47. Cambridge university press, 2018.